

- Q.34 Write short note on lime slaker.
Q.35 Write reactions in oxidation zone in recovery furnace.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain construction and working principle of Lime Kiln with neat diagram.
- Q.37 Calculate the water evaporated from B/L for following data. Solid concentration of weak B/L = 19%, Solid concentration of heavy B/L = 52%, feed rate of weak B/L = 825 Kg/hr.
- Q.38 Write short note on any two of the following
- Cyclone type Evaporator
 - Startup and shutdown procedure of MEE
 - Green liquor Clarifier

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5th Sem / P & P Subject:- Chemical Recovery

Time : 3Hrs. M.M. : 100

SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (10x1=10)
- Q.1 Write main constituent of Green Liquor
- Sodium Hydroxide
 - Sodium Carbonate
 - Sodium Chloride
 - All of the above
- Q.2 What is the unit of Volume
- Kg
 - m/kg
 - Kg/m³
 - m³
- Q.3 Write name of Soda Ash
- Sodium Hydroxide
 - Sodium Chloride
 - Sodium carbonate
 - none of the above
- Q.4 % Sulfidity is _____?
- Na₂S x 100 / AA
 - NaOH x 100 / AA
 - AA x 100 / EA
 - None
- Q.5 Expand BPR
- Blue point Rise
 - Blow point Rise
 - Boiling point Rise
 - none of the above

- Q.6 Choose full form of "TW"
 a) Tensile Strength b) Tension wire
 c) Degree twaddle d) None
- Q.7 The chemical formula for lime mud is _____
 a) CaOH b) CaCO₃
 c) (CaOH)₂ d) None
- Q.8 Write Equivalent weight of Caustic Soda is
 a) 40 b) 80
 c) 60 d) 120
- Q.9 Write Equivalent weight of Sodium Sulfide is
 a) 30 b) 39
 c) 65 d) 90
- Q.10 Concentration of Solids measured in
 a) mole % b) mole fraction
 c) percentage d) percentile

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is Heavy black liquor.
 Q.12 Write any one function of recovery furnace.
 Q.13 Expand MEE
 Q.14 Mention any two terms used in recovery section of paper mill.
 Q.15 What is total alkali.
 Q.16 Write concentration range of strong black liquor.

- Q.17 Mention any one function of cascade type direct contact Evaporator.
 Q.18 Write Slaking reaction.
 Q.19 Name the compound responsible for odor of black liquor.
 Q.20 What is AA

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write principle of black liquor recovery.
 Q.22 Write short note on black liquor oxidizers
 Q.23 Give name of various types of evaporators.
 Q.24 Draw a neat sketch of venture scrubber evaporator.
 Q.25 What is incineration of black liquor.
 Q.26 Write reactions involved in causticizing of green liquor.
 Q.27 Explain working of cascade evaporator in recovery section of pulp and paper industry.
 Q.28 Describe lime mud handling process.
 Q.29 Differentiate soda and sulfate black liquor.
 Q.30 Define Causticizing Efficiency and reduction efficiency.
 Q.31 Convert 245 gram NaOH as such into NaOH as Na₂O
 Q.32 Write the functions of green liquor Clarifier
 Q.33 What is Drag Washer